

DKTE College Training

Agenda

- Advanced SQL
- Data Structures
- Advanced Java

Advanced SQL

Joins in SQL

```
SELECT e.ename, d.dname FROM emps e
INNER JOIN depts d ON e.deptno = d.deptno;
```

```
SELECT e.ename, d.dname FROM emps e
LEFT JOIN depts d ON e.deptno = d.deptno;
```

```
SELECT t1.c1, t2.c2 FROM t1
INNER JOIN t2 ON t1.c1 = t2.c2;
```

```
SELECT t1.c1, t2.c2 FROM t1
LEFT JOIN t2 ON t1.c1 = t2.c2;
```

```
SELECT t1.c1, t2.c2 FROM t1
RIGHT JOIN t2 ON t1.c1 = t2.c2;
```

```
SELECT t1.c1, t2.c2 FROM t1
FULL JOIN t2 ON t1.c1 = t2.c2;
-- Not supported in MySQL
-- You can try it in Oracle/MS-SQL server
```

```
-- self join
SELECT e.ename, m.ename FROM emps e
INNER JOIN emps m ON e.mgr = m.empno;

SELECT e.ename, m.ename mname FROM emps e
LEFT JOIN emps m ON e.mgr = m.empno;
```

```
-- joins with group by
SELECT deptno, COUNT(*) cnt FROM emps
GROUP BY deptno;

SELECT d.dname, COUNT(*) cnt FROM emps e
```

```

INNER JOIN depts d ON e.deptno = d.deptno
GROUP BY d.dname;

SELECT d.dname, COUNT(e.empno) cnt FROM emps e
RIGHT JOIN depts d ON e.deptno = d.deptno
GROUP BY d.dname;

```

Multi-table joins

- emps (empno, ename, mgr, deptno)
- meeting (meetno, topic, venue)
- emp_meeting (meetno, empno)
- depts (deptno, dname)
- addr (empno, tal, dist)
- Relations
 - emps (*) --- (1) depts
 - depts pk -- emps fk
 - emps (1) --- (1) addr
 - emps pk -- addr fk/pk
 - emps () --- () meeting
 - via aux table -- emp_meeting
 - emps pk -- emp_meeting fk
 - meeting pk -- emp_meeting fk

```

-- print ename, tal, dname.
SELECT e.ename, a.tal, d.dname FROM
emps e
INNER JOIN addr a ON e.empno = a.empno
INNER JOIN depts d ON e.deptno = d.deptno;

SELECT e.ename, a.tal, d.dname FROM
emps e
INNER JOIN addr a ON e.empno = a.empno
LEFT JOIN depts d ON e.deptno = d.deptno;

```

```

-- print ename, meeting topic
SELECT e.ename, m.topic FROM emps e
INNER JOIN emp_meeting em ON e.empno = em.empno
INNER JOIN meeting m ON m.meetno = em.meetno;

```

```

-- print ename, meeting topic, addr
SELECT e.ename, m.topic, a.tal FROM emps e
INNER JOIN emp_meeting em ON e.empno = em.empno
INNER JOIN meeting m ON m.meetno = em.meetno
INNER JOIN addr a ON a.empno = e.empno;

```

```
-- print ename, meeting topic, addr, dname
SELECT e.ename, m.topic, a.tal, d.dname FROM emps e
INNER JOIN emp_meeting em ON e.empno = em.empno
INNER JOIN meeting m ON m.meetno = em.meetno
INNER JOIN addr a ON a.empno = e.empno
LEFT JOIN depts d ON e.deptno = d.deptno;
```

Analytical Queries -- WITH ROLLUP

```
SELECT * FROM emp;

SELECT ename, job, sal, deptno FROM emp;

SELECT DISTINCT job FROM emp;

SELECT ename, job, sal, deptno FROM emp
ORDER BY deptno, job;

SELECT deptno, SUM(sal) FROM emp
GROUP BY deptno;

SELECT SUM(sal) FROM emp;
```

```
-- display total sal per dept as well as total sal of all emps.
(SELECT deptno, SUM(sal) total FROM emp
GROUP BY deptno)
UNION
(SELECT NULL, SUM(sal) total FROM emp);

SELECT deptno, SUM(sal) total FROM emp
GROUP BY deptno
WITH ROLLUP;

SELECT job, SUM(sal) total FROM emp
GROUP BY job
WITH ROLLUP;
```

```
-- display total sal per dept per job
SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY deptno, job;

-- display total sal per dept per job -- also show total sal per dept and of all emps
SELECT deptno, job, SUM(sal) total FROM emp
```

```
GROUP BY deptno, job
WITH ROLLUP;

-- display total sal per dept per job -- also show total sal per job and of all
emps
SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY job, deptno
WITH ROLLUP;

-- display total sal per dept per job -- also show total sal per job, total sal
per dept and of all emps
(SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY deptno, job
WITH ROLLUP)
UNION
(SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY job, deptno
WITH ROLLUP);

SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY deptno, job
WITH CUBE;
-- group by -- per dept per job total sal
-- cube -- subtotal by first group by column
-- as well as subtotal by second group by column
-- as well as final total
```

```
-- display per job subtotal and per dept subtotal and final total
SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY deptno, job
WITH ROLLUP;

-- per dept subtotals
SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY deptno, job
WITH ROLLUP
HAVING job IS NULL;

-- per job subtotals
SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY job, deptno
WITH ROLLUP
HAVING deptno IS NULL;

(SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY deptno, job
WITH ROLLUP
HAVING job IS NULL)
UNION
(SELECT deptno, job, SUM(sal) total FROM emp
GROUP BY job, deptno
```

```
WITH ROLLUP  
HAVING deptno IS NULL);
```

```
CREATE TABLE t1 (c1 INT);
```

```
INSERT INTO t1 VALUES (1), (1), (200), (null), (0);
```

```
CREATE TABLE t2 (c2 INT);
```

```
INSERT INTO t2 VALUES (1), (1), (1), (400), (null);
```